

Verifying Efficient Checkpointing Protocol for LLM Training

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Abstract

Large-scale LLM training jobs run across thousands of accelerators for weeks, and link flaps, hardware faults, and software crashes interrupt them regularly. To avoid restarting from scratch, a job periodically checkpoints its state and resumes from the most recent snapshot after a failure. Frequent snapshots bound the work lost to a failure, but each one competes with training for time, so recent systems strive to overlap snapshotting with training. Doing so correctly is hard: training state is spread across servers and across the messages in flight between them, and a snapshot assembled from servers that record at different moments may not correspond to any state the job ever passed through. Such a checkpoint undermines the fault tolerance it was meant to provide, since a job restored from it can silently compute wrong results, and the rare interleavings that cause this are beyond what testing can expose.

We present vCheck, a checkpointing protocol for distributed LLM training that is both efficient and formally verified. vCheck adapts the Chandy-Lamport snapshot algorithm to record checkpoints concurrently with training. Its proof, mechanized in Rocq, guarantees that every checkpoint vCheck produces is a globally consistent cut of the execution, a state the job could genuinely have reached, so restoring from it is sound. We have planned to compare vCheck and the conventional checkpointing protocol used by PyTorch. To our knowledge, vCheck is the first checkpointing implementation for LLM training with a machine-checked proof of correctness. vCheck will be open-sourced.

1 Introduction

Training a modern large language model (LLM) enlists tens of thousands of accelerators that must run in concert for weeks or months [17]. At this scale, failures are the common case, not the exception. Link flaps, GPU and memory faults, and node crashes routinely interrupt long-running jobs: Meta reports 419 unexpected interruptions over a single 54-day run of Llama 3 [12], and other operators report comparable rates [16, 17]. Each interruption discards all computation since the last saved state, turning lost progress into wasted GPU time worth millions of dollars on a single run [1]. *Checkpointing*, which periodically persists a consistent snapshot so a job can resume rather than restart, is the primary defense, and a growing body of work has steadily driven its overhead down [13, 14, 23, 30, 31, 33–35].

Checkpoint overhead matters because every checkpoint-

ing protocol balances checkpoint frequency against training speed. Checkpointing more frequently bounds how much progress will lose after failure, but each checkpoint adds work to the training loop and prolongs the run. Checkpointing less often keeps the run fast, but a failure then rolls the job back to a checkpoint far behind the point of the crash, and all progress in between is lost. Conventional implementations, such as PyTorch’s `torch.save`, make this trade-off especially harsh: they write the checkpoint to persistent storage synchronously and stall the critical path of training for the entire write. To reach a better sweet spot, Recent works such as Checkmate [1] and PyTorch’s experimental feature (`torch.async_save`) [24], decouple checkpointing from the training loop, so that persisting a checkpoint overlaps with computation instead of blocking it. This decoupling buys concurrency, but concurrency is also where bugs can easily hide. In §2 we will show, today’s checkpointing solutions contain numerous bugs that break the correctness guarantee a checkpoint exists to provide, which is a serious concern that even outweighs raw performance.

A checkpointing protocol is only useful if the snapshot it produces is correct. LLM training is a highly concurrent computation that interleaves data, tensor, pipeline, and expert parallelism with collective communication such as ring all-reduce and all-to-all. Capturing a *globally consistent* snapshot of such a system asynchronously, without pausing training, is subtle: a silently inconsistent snapshot is worse than no snapshot at all, because a job that resumes from it may diverge or corrupt its results long after recovery. Yet existing high-performance systems optimize aggressively for speed [13, 23, 24, 31] while resting correctness on testing alone, which cannot rule out the concurrency corner cases those optimizations introduce.

Realizing the drawbacks of existing checkpointing implementations, we aim to develop vCheck, an efficient checkpointing protocol with a machine-checked proof of correctness, and the first formally verified checkpointing implementation for LLM training. It guarantees that every snapshot is a globally consistent cut, i.e., a state that could have arisen at a single point of the run across all servers, so resuming from it faithfully continues the training job.

Main challenges. Building vCheck requires addressing two challenges: (1) *How to checkpoint efficiently?* (2) *How to checkpoint correctly?*

How to checkpoint efficiently? Conventional approach (e.g., `torch.save`) sequentially run saving checkpoint and training computation. This is simple but unacceptable for LLM

training, because the checkpoint persistence is expensive and training would be much blocked if the persistence stays on the critical path. To improve the efficiency, vCheck must instead capture the snapshot asynchronously, overlapping it with on-going training so that steady-state throughput is essentially unaffected.

How to checkpoint correctly? Ruling out the bugs that lurk in optimized checkpointing code means reasoning about every possible interleaving of concurrent training and snapshot messages. Testing and model checking are insufficient here, because they cannot cover the unbounded space of executions a real deployment exhibits [7]. Machine-checked formal verification closes this gap. However, formal verification is known to be complex and challenging. Although it has reached file systems, storage, and consensus [3, 8, 15], no prior checkpointing system has been verified in this way.

Key Ideas. We build vCheck by taking the classical Chandy-Lamport snapshot algorithm [6] as a starting point. The Chandy-Lamport algorithm records a *global cut* of a running distributed system. It does so without ever pausing the system. The algorithm circulates *marker* messages. Each marker splits a channel’s stream into the messages that precede the snapshot and those that follow it. This non-blocking behavior is what makes the algorithm attractive for LLM checkpointing. It is also well proven in the role: the Chandy-Lamport algorithm and its descendants underpin the asynchronous snapshots of stream-processing engines such as Apache Flink [2] and graph-processing systems such as GraphLab [21]. However, today’s LLM checkpointing systems take a different path. They either save and recover each server’s state in isolation [10, 13, 23], or they quiesce training behind a global barrier before dumping state [25, 30, 31].

The Chandy-Lamport algorithm alone, however, is not a checkpointing implementation. It is stated over an abstract distributed system in which a process’s state and a channel’s messages are opaque. It dictates *when* to record state and guarantees a consistent cut, but does not define *what* that state is for a training job or how to capture it quickly. An implementation must also handle what the algorithm abstracts away: the coordination and synchronization among threads on one machine, and the communication among machines. vCheck must therefore capture this implementation logic in a formal specification and prove that its implementation satisfies the specification, so that subtle bugs, such as the concurrency bugs already found in PyTorch’s checkpoint implementation (§2), cannot arise.

Our contributions. vCheck will contribute in three ways.

(1) vCheck implements efficient checkpointing by completely decouple the checkpointing logic from the training workflow. Compared with the conventional approach (e.g., `torch.save`), vCheck avoids stalling the training job during checkpointing. Compared with recent approach (e.g., Checkmate [1]), vCheck does not require special hardware (e.g., programmable switches) support and can be deployed in more

general settings.

(2) vCheck builds its proof atop Grove and targets a single theorem, *exact global recovery*, via an invariant that closes every path to concurrency bugs such as torn reads and premature slot reuse. vCheck’s Go implementation is translated into Rocq statement by statement, and we verify the translated code against the model, so the theorem holds of the code that actually runs rather than of an abstraction of it.

(3) vCheck is the first checkpointing implementation backed by a machine-checked proof of correctness. Every snapshot it records is provably a globally consistent cut of the execution. This delivers a level of assurance that testing alone cannot reach.

2 Background and Motivation

Decoupling design for efficient checkpointing implementation. An efficient checkpointing implementation is pivotal to both training speed and progress preservation. With a costly implementation, frequent checkpointing imposes substantial overhead on training and delays its completion. Infrequent checkpointing, on the other hand, loses more progress whenever the training framework crashes and must recover from the latest checkpoint. PyTorch illustrates the cost. Its conventional checkpointing routine, `torch.save`, writes the model state to persistent storage (e.g., disk) synchronously, blocking the training workflow for the entire duration of the write.

To address this inefficiency, recent work has recognized that checkpointing logic should not be tightly coupled to the training workflow. Instead, decoupling has emerged as the guiding principle for building more efficient checkpointing protocols. Gemini [31], DataStates-LLM [22], and Checkmate [1] all follow this principle. The PyTorch community has moved in the same direction: its recent releases integrate the experimental `torch.async_save` feature, which decouples checkpointing from the training workflow [24].

The benefit of decoupling is evident. The cost of checkpointing is hidden behind the training workflow, so a decoupled design incurs far less overhead and far less stall time than a coupled design that executes checkpointing and training computation in sequence. This benefit, however, does not come for free. The conventional coupled design is inefficient, but by running checkpointing and training one after the other it sidesteps the complexity of concurrency altogether. Decoupling buys its efficiency precisely by introducing concurrency, and concurrency brings correctness risk. Compounding the risk, decoupled designs are still new: they have not been rigorously validated, and most implementations remain research prototypes or experimental features. Indeed, by scrutinizing the open-source implementation of `torch.async_save` in PyTorch, we find it to be bug-prone.

Correctness risks in decoupled checkpointing implementations. Recent experimental feature (`torch.async_save`) removes much overheads due to synchronization, but it still

Defect	What happens
Crash mid-save <i>consistency</i>	Filenames carry rank and counter, never the step. One rank writes step $N+1$, another dies first; save raises, so the operator restores what looks like the surviving checkpoint and loads a model half N , half $N+1$.
Reused staging buffer <i>consistency</i>	Buffers are keyed by source storage, so step $N+1$ stages into the buffer step N is still uploading from.
Dead writer thread <i>completeness</i>	An IO error kills a writer; <code>join()</code> returns normally and nothing compares files written against files requested. <code>save</code> succeeds, <code>load</code> fails.
Round with no manifest <i>completeness</i>	<code>Commit</code> returns with no index and no expected count; <code>load</code> rebuilds each shape from the files present. A finished and a half-finished save are indistinguishable.

Table 1: Defects in PyTorch’s asynchronous checkpointing (`torch.async_save`) and the checkpoint property each one violates.

incurs blocking when checkpoint state is generated at high frequency but the saving is bottlenecked by the storage. More importantly, PyTorch assumes only 1 checkpoint can exist in the same time. When there are multiple checkpoint in-flight (e.g., one checkpoint is being persisted whereas another checkpoint is already generated in memory), PyTorch does not have built-in mechanism to manage such concurrency and relies on the user to handle that [24], which leaves more bug-prone risk to the developer. Table 1 summarizes the defects we found. At a high level, all these bugs are related to the concurrency involved in the training framework where checkpointing and training logic are running asynchronously in parallel. Handling such concurrency is always a challenge to the developers since one careless overlook can lead to a subtle bug that is hidden inside the codebase and may only be revealed after long-term laborious testing.

Formal verification as a fundamental approach to ruling out bugs. The defects in Table 1 are not performance blemishes. Each one can hand a resumed job a state that the original run never passed through, so the job silently trains toward a wrong result. Because that state never existed, the run cannot be reproduced either, even though reproducibility is essential in settings such as model tuning and algorithm analysis. Such bugs, however, can hardly be detected, let alone eliminated, by bounded testing: a test exercises only the interleavings it happens to trigger, and every defect in the table lives in a crash or race that ordinary runs rarely hit. Eliminating them requires an argument that covers every interleaving at once, and that is

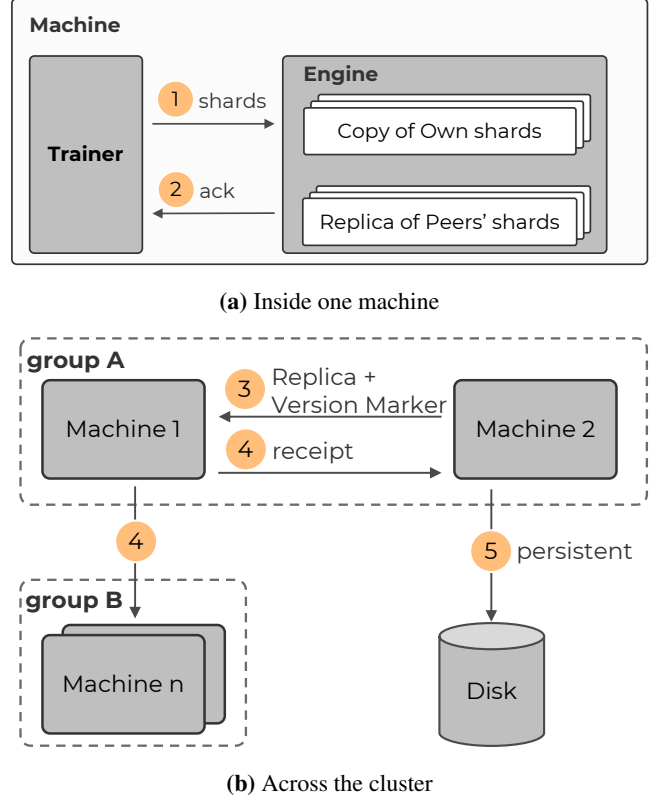


Figure 1: vCheck’s main steps. (a) A version enters the engine (①, ②). (b) It is replicated to its holders (③), acknowledged by receipts (④), and written to disk (⑤). Each engine keeps a copy of its own shard and replicas of its peers’.

precisely what formal verification provides. It establishes the correctness of an implementation through a rigorous proof over all possible executions, and having a machine check that proof closes the last gap, since no case can be skipped and no step taken on faith. Formal verification is therefore a promising path to correct and resilient LLM training systems.

Our motivation. Recent literature has invested extensive effort in designing and implementing efficient checkpointing protocols, yet prior work has left the verification of their correctness unaddressed. Given the numerous bugs that keep being reported, correctness has become as important a requirement for today’s checkpoint implementations as efficiency. This gap motivates vCheck. Like prior work, vCheck aims for highly efficient checkpointing that supports fault tolerance in LLM training. Unlike prior work, and more importantly, it also establishes high assurance of correctness: its implementation is rigorously checked by a proof assistant, leaving no room for hidden bugs.

3 System Design

3.1 Overview

vCheck decouples checkpointing from training, and replication from iteration boundaries. Despite this decoupling, the cluster maintains its global awareness: every checkpointer can independently determine which versions are fully replicated.

Figure 1 illustrates vCheck’s five-step workflow. The workflow involves two roles: the *trainer*, which runs the LLM training workload, and the *checkpointer*, which saves and replicates the training state. vCheck deploys one checkpointer on each machine, and the checkpointers cooperate to replicate multiple *versions* of the trainers’ state.

Definitions. We describe the workflow based on the following definitions.

Version: a checkpoint of the entire job, spanning the state of every trainer, from which all trainers can be recovered.

Partition: the portion of a version generated by one trainer. A version is partitioned across machines, and each partition is handled by the checkpointer on the same machine as the trainer that produced it.

Piece: the unit of transmission. When a checkpointer replicates a partition, it splits the partition into pieces and transmits them according to a predefined schedule.

Bank: a memory buffer that each checkpointer pre-allocates to hold partitions, both those produced by its local trainer and those replicated from other checkpointers. A bank is divided into *slots*, each holding one partition.

The five steps proceed as follows.

① **Trainer: Announcing a partition.** After every k training iterations, the trainer announces its newly computed partition to its local checkpointer by passing a descriptor that holds the partition’s memory location, size, and digest.

② **Checkpointer: Copying and verifying.** The checkpointer copies the memory region into a slot of its own bank and verifies the copy against the digest. Once the copy is complete and verified, it notifies the trainer, which may then reclaim the region.

③ **Checkpointer: Replicating to peers.** The checkpointer streams the partition to the m peer checkpointers assigned to replicate it. Depending on the scheduling policy, the partition may be split into pieces and transmitted over multiple rounds rather than in a single message. When the transmission ends, the checkpointer appends an end-of-version marker to the stream on each channel, which tells the peer that the partition is complete.

④ **Checkpointer: Gossiping partition receipts.** Once a checkpointer receives the marker from a peer, it broadcasts a receipt to all checkpointers, announcing that it holds the partition. By collecting receipts, every checkpointer learns which partitions have been sufficiently replicated. Such a partition is *stable*, and a version is stable once all of its partitions are; each

checkpointer tracks the newest stable version as `stable()`. A stable version can restore the system state after a crash.

⑤ **Checkpointer: Dumping to disk.** Every d training iterations, the checkpointer writes the stable partitions it holds to local disk. This persistence guards against extreme failures, such as a cluster-wide power loss that wipes every replica in memory: after such a failure, the system can still recover from the most recently persisted version on disk.

3.2 Memory Copy and Replication ①–③

Steps ① and ② keep checkpointing logic entirely out of the training loop. The trainer passes only a descriptor, which records the memory location, size, and digest of its newly computed partition. The checkpointer then reads and copies the partition into its bank without blocking the trainer’s computation.

To prevent *torn reads*, where the checkpointer copies a partition while the trainer is overwriting it for the next iteration, vCheck treats the descriptor as a borrow of the region: the checkpointer holds read-only access, and the trainer must not reclaim the region until the checkpointer returns it. The borrow is the only synchronization point between the two roles. Users can configure the trainer either to wait for the return or to drop the pending checkpoint and keep training without interruption. Crucially, even in the blocking mode, the trainer waits only for a local memory copy, never for the network.

To guarantee data integrity without further synchronization, the checkpointer validates the copied bytes against the digest in the descriptor. Any mismatched partition is discarded, so corrupted data never enters the replication pipeline and the checkpointer’s copy always matches the trainer’s output exactly. Furthermore, vCheck handles the validated partition with no intermediate buffering: the checkpointer copies the bytes directly into a slot of its bank and uses that same slot both to stream the partition to peers for replication.

3.3 Gossip Partition Receipts ④

Because vCheck decouples checkpointing from training, no global barrier stops the trainers while the checkpointers finish replicating a version. A checkpointer may therefore receive pieces that belong to different versions from the same peer. vCheck sends the pieces over in-order network channels, so a channel carries a peer’s pieces version after version, and the end-of-version markers delimit them. Using the markers, the receiver assigns each piece to its version and assembles the pieces into the complete partition for that version.

However, barrier-free replication raises a memory-management challenge: *premature slot reuse*. In vCheck implementation, each checkpointer’s bank is a ring buffer of W slots. A fast checkpointer can fill its bank while a slow peer is still receiving an older partition, and if it overwrites a slot prematurely, it may destroy a partition that belongs to the newest globally stable version. Before reusing a slot, a

checkpoint must therefore confirm that a newer, globally stable version exists across all peers, i.e., the new version is already stable. vCheck establishes this through gossip, by which checkpointers share their replication progress.

Once a checkpoint has received a partition and its end-of-version marker, it broadcasts a receipt to all other checkpointers. From the gossiped receipts, each checkpoint derives which versions have been fully replicated and marks them as stable, advancing a `stable()` watermark that records the newest stable version. A checkpoint overwrites a bank slot only if the partition in that slot belongs to a version below the watermark. Otherwise, it blocks until the watermark advances past that version.

3.4 Dumping to Disk ⑤

To survive catastrophic failures in which every checkpoint replicating the partition has crashed, vCheck maintains a second tier of persistence on disk. Traditional disk checkpoints require cluster-wide coordination: each rank writes its state, a global barrier confirms that all writes have succeeded, and only then is the checkpoint declared valid. vCheck instead decouples disk I/O from cluster coordination entirely. Checkpointers independently write the partitions they hold to local storage, one standalone file per partition, consisting of a header that records the partition identifier, the version number, and the length, followed by the raw bytes.

The disk path follows a copy-then-write pattern. To write a file, the checkpoint pins the corresponding bank slot and copies its contents into a dedicated buffer reserved for disk checkpoints. A separate writer thread then streams the buffer to disk in parallel. While the write is in progress, the trainer, the network transport, and the gossip mechanism all continue without interruption.

3.5 Recovery from Checkpoints

To recover the system state based on checkpoint, vCheck takes two steps. The protocol first attempts a fast recovery from memory: it queries every reachable checkpoint for a target version v , and each checkpoint replies with the validated partitions it holds for t together with its local `stable()` watermark. If t is older than a checkpoint’s watermark, the checkpoint rejects the request and returns its latest stable version, then the protocol restarts the query at that higher version. Once the replies are collected, the protocol assembles a complete version from one replica of each partition.

If the entire cluster has crashed and all checkpoints have been erased from memory, the protocol falls back to the disk tier. It scans the storage directory and checks each file. A file is complete if its header parses and its physical length matches the length recorded in the header; a file cut short by a crash fails this check and is discarded. The tool then picks the newest version that has a complete file for every partition and restores it.

4 Verification Strategy

The goal of our proof is *exact global recovery*: after a failure, vCheck restores a correct version from memory or from disk. In memory, as long as no partition loses all of its replicas, the live checkpointers together hold a correct version that is no older than any stable version they report. On disk, whenever every partition of a version has a complete file, those files form a correct version.

The proof has two halves. First, we model the cluster as a state machine. A *configuration* consists of each checkpoint’s bank and receipts and the messages sent on each channel. We then define five *steps* that move the machine from one configuration to the next. Each step corresponds to an atomic action of a checkpoint. The theorem is stated over this model (§4.2) and is proved through a global invariant that every step preserves (§4.3). Second, using Grove [27] and its Goose front end [4], we prove that every operation of the Go code moves the corresponding configuration by exactly one step, so every run of the code ends in a reachable configuration and the theorem applies to it (§4.4). The proof rests on six assumptions (§4.5).

4.1 Correctness Criterion

To state our goal precisely, we first define when a version is correct.

Definition (correct version). A version is *correct* if it is *complete*, containing a partition for every trainer, and *consistent*, with every partition identical to what its trainer produced when the version was taken.

Both conditions are necessary. A missing partition leaves a trainer nothing to restore, and a partition that mixes data from different iterations yields a state the training job never reached; each defect in Table 1 breaks one of these conditions.

The two conditions are also sufficient. A version is taken at the end of an iteration, when no messages between trainers are in flight, so the state of the job at that point consists only of the states of its trainers. Completeness ensures that the version holds a partition for every trainer. Consistency ensures that each partition is exactly its trainer’s state at that iteration. Together, a correct version is the state of the whole job at that iteration, and restoring it resumes the job as if it had paused there.

vCheck gains its efficiency by removing synchronization, and each removal opens a way to break these conditions. For example, torn reads (§3.2) can make a version inconsistent, and premature slot reuse (§3.3) can make it incomplete. These errors arise only under rare interleavings that testing does not reliably reach, so we prove that no interleaving yields an incorrect version.

4.2 Formal Model and Theorem

The trainer. The trainer is tens of thousands of lines of code and evolves constantly, so we leave it out of the model. The decoupling design of §3.2 makes this possible: the trainer and the checkpointers share nothing but a descriptor. We model the trainer only by its output. We write $oracle(p, v)$ for the state of partition p 's trainer when version v was taken, and assume that the descriptor for (p, v) carries the digest of $oracle(p, v)$ (§4.5).

Configurations. A configuration consists of each checkpoint's bank and receipt matrix and each channel's message log. The model abstracts the bank (§3.1) as a set of slots, each holding one partition at one version together with three flags: whether the slot is *validated*, meaning that the checkpoint treats it as a finished copy that it may replicate, acknowledge, and write to disk; whether it is pinned for a disk write; and whether it has been written to disk. The receipt matrix holds the receipts that a checkpoint has recorded, and $stable()$ is computed from it. A channel's log lists every message sent on the channel, in order, and counts how many the receiver has taken.

Steps. Each step is one operation that a checkpoint performs while holding its lock. **COPY** copies the partition that the trainer announced into a slot, and marks the slot as validated only if the digest of the copy matches the descriptor. **SEND** appends one message to a channel. A message is of one of three kinds: a data message carries a piece of a partition, a marker signals that all of a partition's data for a version has been sent, and a gossip message carries receipts. **RECEIVE** takes the next message from a channel and handles it according to its kind: it writes a data message's piece into the slot for that partition and version, marks that slot as validated upon a marker, and adds a gossip message's receipts to the receiver's receipt matrix. Before writing a slot to disk, the checkpoint copies it into a disk buffer. **PIN** marks a validated slot as pinned before this copy, and no step may reuse a pinned slot. **UNPIN** clears the mark after the copy and marks the slot as written.

The model leaves the sending schedule free, so the theorem covers any way of splitting a version into messages and any timing of sends. It imposes obligations only on what each kind of message may carry and when. A data message must come from a validated slot, so no replica holds data the trainer never produced, and must not precede data of an older version on its channel, so late data never lands in a reused slot. A marker must follow all of its version's data, so no half-filled replica is validated. A gossip message may report only receipts that the sender has recorded, so $stable()$ never overstates replication.

Theorem (exact global recovery). Let C be a *reachable* configuration, one that finitely many steps lead to from the empty configuration, and let L be any set of checkpointers that survive a failure.

Memory layer. Suppose every partition has one of its holders in L , and some checkpoint in L records $stable() = v > 0$. Then L holds a correct version $V \geq v$. That is, every partition p has a validated copy of (p, V) at a holder in L (complete), and every validated copy of (p, V) in L equals $oracle(p, V)$ (consistent).

Disk layer. If every partition has a file written for version v , those files form a correct version v .

Recovery. If the recovery tool assembles version v from the replies of checkpointers in L or from files, the assembled version is correct, and v is at least the $stable()$ in every reply it used.

4.3 The Global Invariant

Since a reachable configuration is produced by finitely many steps, we prove the theorem by induction over steps. The theorem itself, however, is not inductive: knowing that a configuration satisfies the theorem is not enough to show that the next one does. Whether a marker validates a correct replica depends on what the channel carried before it, and whether reusing a slot is safe depends on whether the receipts behind $stable()$ are honest; the theorem says nothing about either. We therefore prove a stronger, inductive property instead: the global invariant. It holds in the empty configuration, each of the five steps preserves it, and it implies the theorem, so every reachable configuration satisfies both.

The invariant has four clauses. Two ensure consistency: validated slots hold the trainer's state, and files hold validated data. The other two ensure completeness: receipts are honest, and the newest stable version is never overwritten. We arrived at these clauses while attempting the proof, and stating them exposed bugs in our code (§??).

Validated slots hold the trainer's state. A slot becomes validated either through **COPY** or through **RECEIVE**. **COPY** marks a slot as validated only when the digest of the copy matches the descriptor, which carries the digest of the trainer's state. Assuming an accidental digest collision, a copy damaged by a torn read cannot match. **RECEIVE** marks a slot as validated only upon a marker, and the obligations on data messages and markers ensure that all data before the marker came from a validated slot. Either way, a validated slot holds the trainer's state.

Files hold validated data. **PIN** applies only to a validated slot, and while a slot is pinned, no step can reuse it. The checkpoint copies the slot into its disk buffer during this window and writes the file from that buffer. So every file holds the data of a validated slot, for the partition and version it was written for.

Receipts are honest. A holder sends a receipt for (p, v) only after it has validated its copy of (p, v) , and by the obligation on gossip, a gossip message reports only receipts that the sender has recorded. So a receipt in any receipt matrix means that its holder validated that copy.

The newest stable version is never overwritten. Let V be the largest `stable()` that any checkpointer has recorded. `stable()` reaches V only when the receipt matrix holds a receipt for V from every holder of every partition, so by the previous clause every holder has validated its copy of V . A holder reuses a slot only if the slot’s version is below its own `stable()`, which is at most V . So no holder ever reuses its copy of V .

Finally, the invariant implies the theorem. For the memory layer, the newest stable version V is never overwritten, so every partition keeps a validated copy of V at its holder in L , and every validated slot holds the trainer’s state; hence L holds a correct version $V \geq v$. For the disk layer, every file holds validated data, so files for every partition of v form a correct version v . For recovery, the tool uses only validated copies and complete files, and it asks again whenever a reply reports a `stable()` above v , so the version it assembles is correct and no older than any `stable()` it saw.

4.4 Connecting the Model to the Code

We map the Go state of all checkpointers, together with the messages in flight, to a configuration, and prove that every state-changing operation moves it by exactly one step. Each such operation runs under the checkpointers’s lock, so it takes effect atomically, and a global log of the steps taken so far lets induction show that the mapped configuration is always reachable. For COPY, PIN, and UNPIN, which touch only local state, the mapping is direct. SEND and RECEIVE are harder, because they also touch the channel between two checkpointers.

Receive. The model’s RECEIVE takes the next message on a channel, but a real network may drop, duplicate, or reorder messages. We prove that vCheck’s transport nonetheless delivers each channel’s messages in the order they were sent, with none skipped or repeated, so every receive in the code is a RECEIVE in the model.

Send. The model’s SEND accepts only messages that meet the obligations of §4.2. The sending code, however, chooses its own schedule, and nothing in it enforces these obligations by construction. The proof therefore keeps a global record of what each channel has carried so far, and checks each new message against this record and against the sender’s own state. Every send in the code thus meets the obligations and is a SEND in the model.

4.5 Trusted Assumptions

The theorem holds of a running cluster under the following assumptions. (1) *Trainer side.* The host-memory copy returns the data of the announced region and runs at most once per announcement; each announcement is a fixed byte string; and each descriptor for a version newer than any earlier descriptor carries the digest of $oracle(p, v)$. (2) *Digest.* FNV-1a-128 has no accidental collision among the payloads of a run. (3) *Grove primitives.* Grove’s network send and file write meet

Layer	Lines
Code proofs	22,768
Transport engine	3,213
Model, invariant, and preservation	28,347
Protocol mathematics	254
Trace coupling and recovery	2,726
Top-level claims and closed run	2,316
Total	60,926
Proof-to-code ratio	16.3×

Table 2: Lines of proof by layer, excluding blanks and comments, and the ratio of total proof lines to the 3,742 lines of Go.

their specifications, and every run starts in a fresh directory, so no file of the run exists at boot. (4) *Deployment.* Every checkpointers starts with the same flags, which are checked at startup, and runs once, with a single sender that sends only through the verified send paths. (5) *Arithmetic.* The proof covers only executions without integer overflow; at runtime, an overflow panics and stops the checkpointers instead of producing a wrong value. (6) *Foundations.* The Rocq kernel, Iris, Perennial, Goose, and classical logic are sound.

5 Proof Effort

vCheck’s verified checkpointers is 2,173 lines of Go. The trusted code around it, namely the copy primitive, the control channel, the two daemons, and the recovery tool, is another 1,569 lines of Go, plus a 990-line PyTorch trainer. Table 2 breaks down the proof, in lines of Rocq, by layer.

The first two rows, which prove the Go code against the model, amount to about 16.3 lines of proof per line of code, within the range reported by Grove [27], vMVCC [7], and GoJournal [5].

6 Planned Evaluation

Because of its decoupled design, vCheck is implemented as a pluggable module that works with existing training frameworks with minimal integration effort. We are currently connecting vCheck to PyTorch through shared memory, so that vCheck can checkpoint and replicate the state of every node in an LLM training job. We then plan to compare vCheck against PyTorch’s built-in checkpointing mechanism and measure how each implementation affects end-to-end training time.

7 Related Work

Fault-tolerant LLM training. Large-scale LLM training must tolerate a variety of failures in practical deployments. Existing work falls into two classes: checkpoint-based and checkpoint-free approaches. Checkpoint-free approaches tolerate failures

by exploiting the redundancy that already exists in the training job. Bamboo [29], for instance, has each node in a pipeline redundantly compute the layers of its neighbor, so that when a node is preempted its neighbor already holds the state needed to carry on. ReCycle [11] exploits data parallelism, in which every pipeline stage has peers holding identical parameters. When a node fails, ReCycle reroutes its micro-batches to those peers, fitting the rerouted work into existing pipeline gaps rather than stalling the pipeline. FleetIQ [9] exploits the rich connectivity of LLM training networks. When a NIC or a link flaps, it reroutes traffic over healthy links and NICs, masking the failure from the training framework above. Checkpoint-free approaches are attractive, but each applies only to specific workloads, failure types, or deployment settings. Fault tolerance for general large-scale LLM training therefore still requires checkpoint-based approaches, which have drawn numerous efforts in the recent literature, as we describe next.

Efficient checkpointing implementation. Prior work reduces the overhead of checkpointing so that checkpoints can be taken frequently. One line overlaps persistence with training. CheckFreq [23] pipelines the snapshot of model state with the next iteration and tunes the checkpoint frequency online. DataStates-LLM [22] exploits the fact that model and optimizer shards remain immutable for long stretches of an iteration and lazily copies them to host memory, and then to storage, in the background. PyTorch’s asynchronous distributed checkpointing adopts the same staging pattern [24], and ByteCheckpoint [30] adds a parallelism-agnostic checkpoint format with load-time resharding and full-stack I/O optimizations. A second line keeps checkpoints in memory and relies on replication to tolerate crash failures. Gemini [31], PHOENIX [34], Swift [35], and Just-In-Time Checkpointing [13] all keep each node’s state in the memory of its peers, either by replicating it explicitly or by exploiting the replicas that data parallelism already maintains, and fetch that state to recover when a node crashes. Finally, Checkmate [1] leverages programmable switches to mirror gradients in flight to shadow servers, which apply them to reconstruct the checkpoint state. Checkmate adds negligible overhead to training, but it depends on special hardware and is therefore limited in where it can be deployed. vCheck’s implementation is inspired by these works and incorporates both the decoupling design and the replication approach to checkpoint efficiently. Unlike prior work, however, vCheck aims for both high performance and high fidelity in its checkpointing implementation. Whereas all prior systems optimize for speed and validate their designs by testing alone, vCheck establishes its correctness with a machine-checked formal proof.

Formal verification framework. vCheck builds on a line of foundational verification frameworks developed in Rocq (formerly Coq) [28]. Iris [18] provides a higher-order concurrent separation logic. Perennial [3] extends Iris with crash-safety reasoning and, through Goose [4], applies it to Go programs.

Grove [26, 27] further adds distributed-systems primitives such as RPCs, replicated state machines, and leases. We build vCheck on Grove for its mature separation-logic support and rich tactic library. vCheck’s proof methodology, however, is not tied to Grove. In principle, it applies equally to other verification frameworks, including Dafny [20], IronFleet [15], Verdi [32], and Verus [19].

8 Conclusion

We have presented vCheck, an efficient checkpointing protocol that provides fault tolerance for LLM training without slowing it down. vCheck decouples checkpointing from the training workflow, so training never blocks on a checkpoint in progress. Building on Grove, we develop a machine-checked proof for vCheck, to demonstrate that every checkpoint it produces is a state the training job could genuinely have reached, so resuming from it is sound. vCheck thus offers both high efficiency and high fidelity in checkpointing. We plan to evaluate vCheck against conventional checkpointing implementations in the near future, and we will open-source both its implementation and its proof.

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